NUMERICAL INVESTIGATION OF A LIQUID DROPLET TRANSPORTED BY A GAS STREAM IMPINGING ON A HEATED SURFACE: SINGLE-PHASE REGIME

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ABSTRACT

Gas-assisted sprays, in which a gas stream propels droplets to a target surface, have been demonstrated to be more efficient in heat removal than sprays generated solely by liquid droplets. In this paper a numerical model, using the volume of fluid (VOF) method and an adaptive grid technique, is developed to investigate the droplet spreading dynamics and heat transfer due to a single drop that is transported by a gas stream onto a heated surface. The results show a significant deviation from the behavior of a free falling droplet that is generated at the same height due to the pronounced effects that the carrier gas stream has on the spreading rate, liquid film thickness and surface temperature distribution. The gas stream transports the liquid droplets to the surface imposing pressure and shear stress over its surface, which accelerates the droplet, increases the maximum spreading diameter, and decreases the liquid film thickness at the moment of maximum spreading. This contributes to an increase in the wetted area where heat transfer takes place. Furthermore, advection within the droplet during the early stage of impact deviates the temperature distribution inside the liquid droplet from a pure conduction problem. The numerical computations, which have been undertaken assuming laminar behavior, reveal that the gas flow in the vicinity of the droplet is quite complex. Surprisingly, the laminar computations agree well with experiments even when the impinging gas stream is known to be turbulent.

KEY WORDS: Spray cooling, gas-assisted spray cooling, propelled droplet, droplet impact, VOF

NOMENCLATURE

Cp	heat capacity, J/kg-K
D	diameter, mm
F	volumetric force due to surface tension, N/m^3
H	nozzle-to-plate distance, mm
P	pressure, Pa
Re	Reynolds number
T	temperature, ˚C
We	Weber number
b	physical and thermodynamic properties
g	gravity, m/s^2
h	heat transfer coefficient, W/m^2-K
k	thermal conductivity, W/m-K
$\hat{n}$	surface normal
q''	heat flux, W/m^2
q'''	heat generation, W/m^3
t	time, ms
v	velocity, m/s

Greek Symbols

Γ	stress tensor
α	volume fraction
μ	viscosity (kg/m-s)
θ	contact angle (º)
ρ	density (kg/m^3)
σ	surface tension (N/m)
τ	non-dimensional time
ξ	spreading factor

Subscripts

a	ambient
air	air
foil	stainless steel foil
jet	impinging jet
o	droplet
w	water
wall	wall
∞	ambient

INTRODUCTION

Throughout the years, proper heat removal has been one of the most important aspects in the reliability of the new technologies that require thermal management. One technique to remove heat, which has gained wide attention due to its capacity to extract heat from a surface more efficiently and homogeneously, is spray cooling. To study the fundamentals of this problem, many researchers have focused on the behavior of a free falling droplet impinging onto a flat surface.

[1-11]. It has been demonstrated that the deformation process of a free falling droplet impinging on a surface is governed mainly by impact velocity, initial droplet diameter, surface tension effects, and wall surface temperature. For this reason, the specification of the proper initial and boundary conditions is fundamental to accurately modeling this phenomenon, either analytically or numerically. Experimentally, the main problem resides in the fast and precise measurement of the wall temperature drop necessary to capture the early stage of the impact. Previous studies have revealed that temperature decay over the heated wall occurs in a range of a few milliseconds [12]. Many analytical [13-16] and numerical [17-19] formulations have been developed to predict the evolution of the drop diameter during the deformation process. Harlow and Shannon [20] first solved the flow dynamics during the splash of a liquid droplet onto a flat plate using the Marker-and-Cell technique. Since its introduction, the volume of fluid (VOF) method has gained wide acceptance to characterize the interaction at the liquid-gas interface and investigate the dynamics of a liquid droplet spreading and receding on a flat surface [21, 22]. However, its greatest disadvantage resides in the calculation of the liquid-gas interface at the solid surface, where the contact angle has to be empirically specified.

In investigations related to heat transfer, most of the studies have considered evaporation of liquid droplets that are gently deposited on heated horizontal surfaces [23-28]. Chandra *et al.* [23] studied the effect of liquid-solid contact angle on droplet evaporation. It was shown that heat transfer can be enhanced by reducing the contact angle, which increases the liquid-solid contact area and therefore, the evaporation rate. Di Marzo *et al*. [25] studied the transient thermal behavior of a single water droplet that was deposited over the surface of a semi-infinite solid. In their study two models for the prediction of the thermal behavior of the droplet-solid interaction during evaporative cooling were proposed. The first one solved the liquid and solid transient conduction equations simultaneously, and the second one introduced a constant and uniform heat flux boundary condition at the liquid and solid interface. Ruiz and Black [26] performed numerical studies of the evaporation process of small water droplets on hot solid surfaces taken into account the internal fluid motion that occurs as a result of the thermocapillary convection in the droplets and its effect on the heat transfer between the drop and the solid surface. Other studies have investigated the evaporative cooling by the impingement of single water droplets [29-37]. However, few studies have investigated the heat transfer during droplet impact due to convection and conduction in single-phase. Healy *et al.* [38] studied the effect of heat transfer on the spreading process to validate the assumption that the droplet spreads adiabatically over the surface. The VOF method and local grid refinement technique have also been implemented to model the fluid mechanics and heat transfer during droplet impact [29, 30, 39]. Pasandideh-Fard *et al.* [40] performed both experiments and numerical simulations using the VOF model to measure the temperature distribution within a water droplet. They also developed an analytical model to estimate the droplet cooling effectiveness.

Another technique that has been shown to be more efficient in removing heat than spray cooling alone is gas-assisted spray cooling. In contrast with the conventional spray systems, droplets are transported to a target surface by a gas stream, which accelerates them, imposes pressure and shear stress over their surface, enlarges the maximum spreading diameter, and decreases the liquid film thickness [41]. In addition, if evaporation takes place, the vapor can be removed more easily from the surface by the gas stream. Nevertheless, no fundamental work has been done in this area. In this paper a fundamental study of a gas-assisted spray cooling problem is presented, where a single droplet is accelerated and transported by a gas stream toward a heated surface.

EXPERIMENTAL APPARATUS AND PROCEDURE

The experiment shown in Fig. 1, was designed to establish an idealized situation of a droplet transported to a target surface by a uniform flow. The surface was a 25.4 mm diameter and 0.0254 mm thick stainless steel foil. To create the flow, an axisymmetric converging nozzle with a 36:1 area ratio was utilized. The nozzle exit diameter (D_{jet}) was 25.4 mm and it was positioned so that H/D_{jet} was 2. Measurements of the air velocity at the exit plane were performed by neglecting the friction of the air flowing through the nozzle and solving the conservation of mechanical energy along a streamline. Then,

$$v_{jet} = \sqrt{2\frac{\Delta P}{\rho_{air}}} \tag{1}$$

where ρ_{air} is the density of air and ΔP is the pressure difference between the nozzle inlet and nozzle exit. The pressure difference was acquired with a MKS instrument 698A Baratron, which assumed that the nozzle exit was under atmospheric conditions. Water droplets were generated from a hypodermic needle, which was placed 102 mm above the jet exit. High speed digital video images were captured with a Phantom V7.1™ color camera at a nominal rate of 16000 fps with an exposure time of 59 μs.

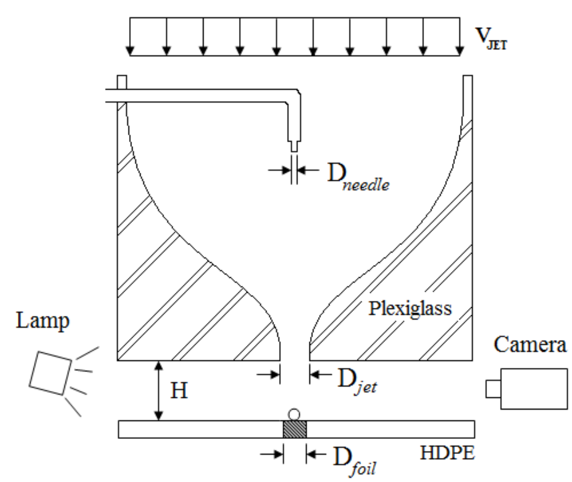

Fig. 1 Experimental Apparatus Schematic

MATHEMATICAL MODEL

The computational simulations were carried out using a two-dimensional axisymmetric laminar model in FLUENT for the ANSYS release version 12.0.1. The VOF model was implemented to track the liquid-gas interface, where a continuity equation for the secondary-phase (water) volume fraction was solved:

$$\frac{\partial \alpha_2}{\partial t} + \vec{v} \cdot \nabla \alpha_2 = 0 \tag{2}$$

The primary-phase (air) volume fraction was computed based on Eq. (3)

$$\alpha_1 = 1 - \alpha_2 \tag{3}$$

Surface tension was included in the momentum equation, Eq.(4), as a source term ($\vec{F}$)

$$\rho\left(\frac{\partial \vec{v}}{\partial t} + \vec{v} \cdot \nabla \vec{v}\right) = -\nabla p + \nabla \cdot \vec{\Gamma} + \rho \vec{g} + \vec{F} \tag{4}$$

and wall adhesion was incorporated as boundary condition at the solid-fluid interface by the contact angle, which is related to the surface normal as shown in Eq. (5)

$$\hat{n} = \hat{n}_w \cos\theta_w + \hat{t}_w \sin\theta_w \tag{5}$$

where n_w and t_w are the unit vectors normal and tangential to the wall, respectively. The contact angle was extracted from the experiments and it was assumed to be constant during the spreading process. The initial condition was set to be at the instant when the water droplet first touches the surface. Heat transfer within the fluid and solid was obtained by solving the energy equation:

$$\rho C_p \frac{DT}{Dt} = k\nabla^2 T + \dot{q}''' \tag{6}$$

where $\dot{q}'''$ is the constant heat generation within the solid. Initially, the water droplet was assumed to be in thermal equilibrium with its surroundings at a temperature of 22˚C, whereas the surface had an initial temperature of 70˚C. The physical and thermodynamics properties from Eq. (4) and (6), were defined in each control volume as

$$b = \alpha_2 b_2 + (1 - \alpha_2) b_1 \tag{7}$$

Figure 2 shows a schematic of the boundary conditions employed in the simulations. A grid refinement technique was implemented to improve the accuracy of the simulations at the liquid-gas interface, ensuring that a grid independent solution was achieved. Approximately 82000 cells were used for 3 levels of grid refinement (Fig. 3). First-order implicit formulation was used for time discretization, where the time step needed for carrying out the simulation without affecting the results was 1×10^{-6} s. An implicit body force formulation was used in the multiphase model, where the face fluxes for the VOF model, were calculated using the geometric reconstruction scheme.

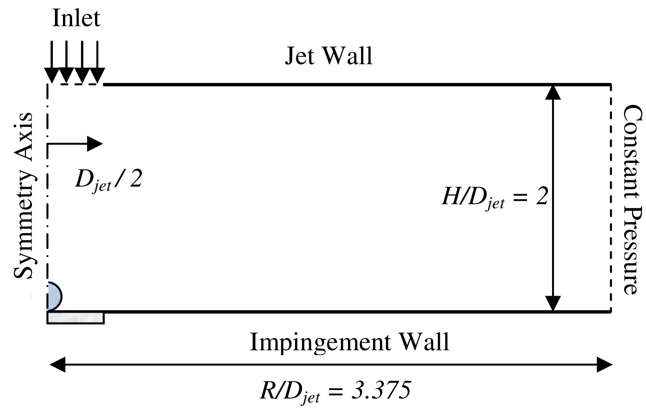

Fig. 2 Computational domain and boundary conditions

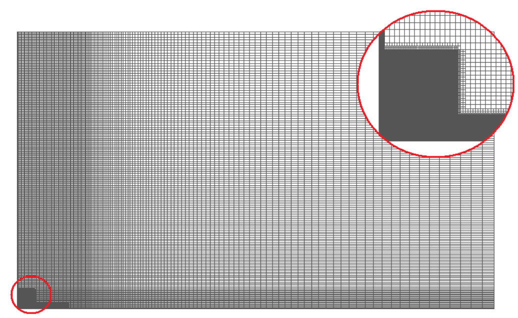

Fig. 3 2-D axisymmetric computational mesh

RESULTS AND DICUSSION

High-speed images were used to measure the parameters required for the simulations such as droplet diameter, impact velocity and contact angle. The measured contact angle was 110 ± 10˚ and the droplet initial diameter was 3.0 ± 0.06 mm. In the simulations both contact angle and droplet initial diameter were held constant at 110º and 3.0 mm, respectively.

Free Falling Droplet

A single droplet event under free fall was studied to investigate the correlation between the droplet dynamics and instantaneous heat transfer. Table 1 shows the different velocities that were achieved in this experiment

Table 1 Droplet impact velocities in free fall

Case	v_o *(m/s)*	Re_o	*We*
1.1	1.46	4579	88
1.2	1.70	5332	119
1.3	2.00	6273	165
1.4	2.27	7119	213

Figure 4 shows the variation in time of the spreading factor at different impact velocities. As seen in previous studies [12, 41] any increase in the inertial forces leads to a increase in maximum spreading diameter. Excellent agreement between the numerical and experimental results was obtained.

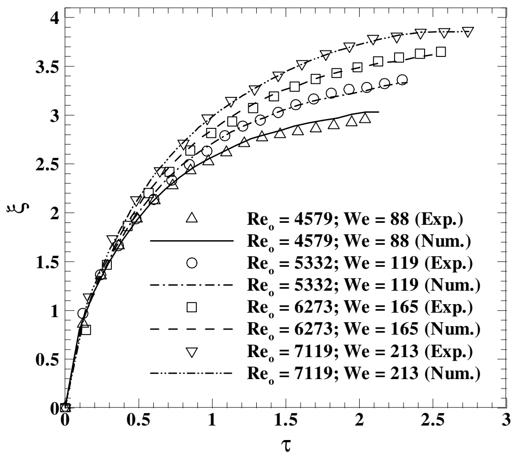

Fig. 4 Spreading factor as a function of time (Free fall)

Immediately after the droplet impacts the surface, a considerable amount of heat starts to be transferred from the hot wall to the liquid droplet, and its magnitude decreases as the droplet spreads. Figure 5 shows the heat flux variation at the wall center.

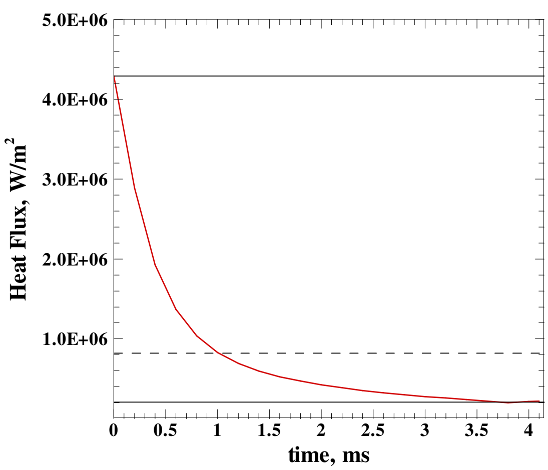

Fig. 5 Heat flux variation at the wall center for Re$_o$ = 5332

In order to investigate the conjugate nature of the problem during the spreading process, the results were compared to the solution of a droplet impinging over a surface that dissipates constant and uniform heat flux, meaning the foil thickness was negligible. The value of the heat flux was assumed to be the initial value at the wall center obtained for the conjugate solution. In addition, the results were also compared to the semi-infinite solid solution for a constant heat flux boundary condition, where the droplet temperature variation was calculated at the stagnation point (z = 0) as shown in Eq. (8)

$$T(t) = \frac{2q''}{k_w}\sqrt{\frac{k}{\rho C_p}\frac{t}{\pi}} + T_\infty \tag{8}$$

Figure 6a shows a comparison of the droplet temperature variation at the point of impact for a constant heat flux boundary condition, conjugate solution, and semi-infinite solid approximation. A significant deviation is observed among the three approaches. It is believed that these differences are consequence of the advection that occurs within the droplet during the early stage of the impact, which deviate the thermal behavior from a pure diffusion problem. In addition, Fig. 6b shows the droplet temperature variation for the conjugate problem compared to the semi-infinite solid solution at three different heat fluxes: the maximum, mean and minimum heat flux achieved during the process. As the magnitude of the heat flux decreases, the semi-infinite solid approximation moves toward the conjugate solution. However, the analytical solution is incapable of properly capturing the temperature response at the early stage, which indicates that the conjugate nature of the problem cannot be neglected and also that heat is mainly transferred by convection rather than diffusion at this stage.

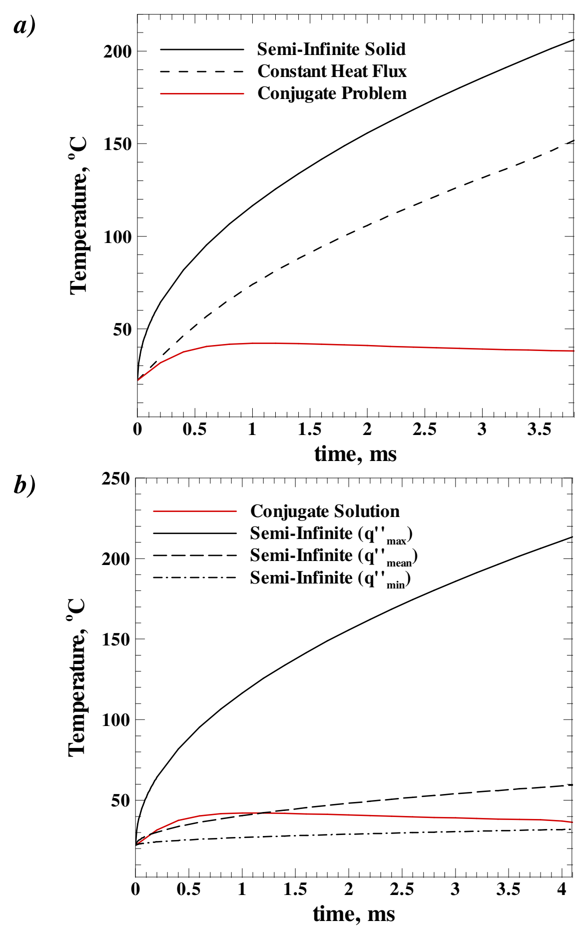

Fig. 6 Droplet temperature variation at the stagnation point for *a)* different solutions and *b)* different heat fluxes (Re$_o$ = 5332)

Propelled Droplet

In this section, four different cases were studied to evaluate the effect of a gas stream in the fluid dynamics and heat transfer. The liquid droplet were generated at a fixed height and the jet velocity was varied from 0 (free fall) to 10 m/s, as shown in table 2, range in which the turbulence nature does not significantly affect the fluid dynamics and then, the laminar assumption can still be applied

Table 2 Jet and droplet impact velocities

Case	v_{jet}	v_o	Re_{jet}	Re_o	*We*
2.1	0.0	1.46	0	4579	88
2.2	5.0	1.55	8191	4861	99
2.3	7.5	1.72	12287	5394	122
2.4	10.0	2.00	16382	6273	165

Figure 7 shows a side by side comparison of the images obtained for Re_{jet} = 16382 between the experiments and the simulations. Regardless of the instabilities seen in the advancing ring, the 2D axisymmetric assumption was very accurate. Less than 4% difference was achieved in the simulations, based on the difference between the experimental and numerical instantaneous droplet diameter.

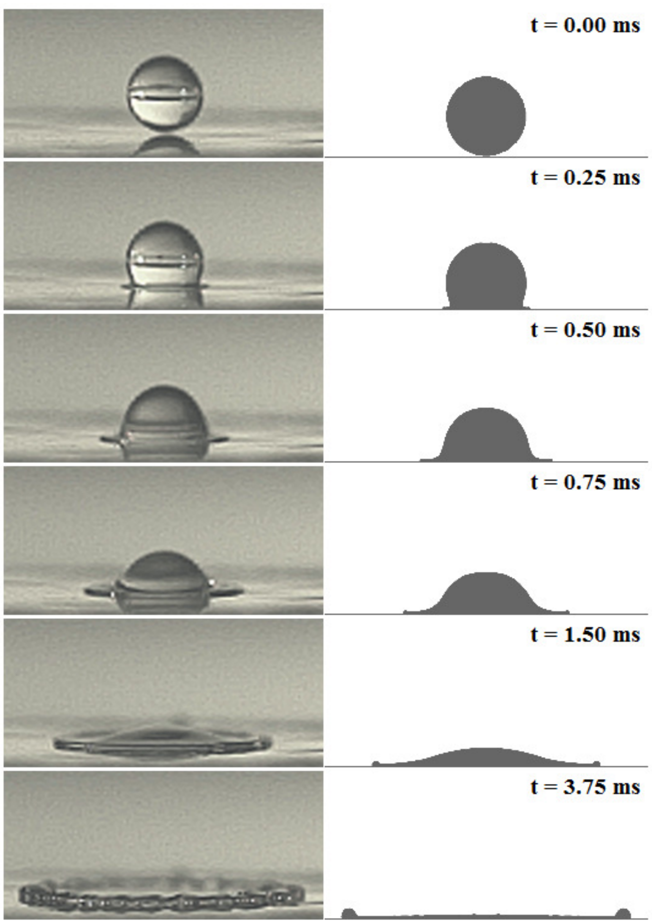

Fig. 7 Comparison between experimental and numerical images obtained for Re_{jet} = 16382

Again, in Fig. 8 the variation in time of the spreading factor is displayed. It can be observed that as the jet velocity increases the droplet impact velocity increases as well. Excellent agreement between the numerical and experimental results was obtained even when the flow was assumed to be laminar.

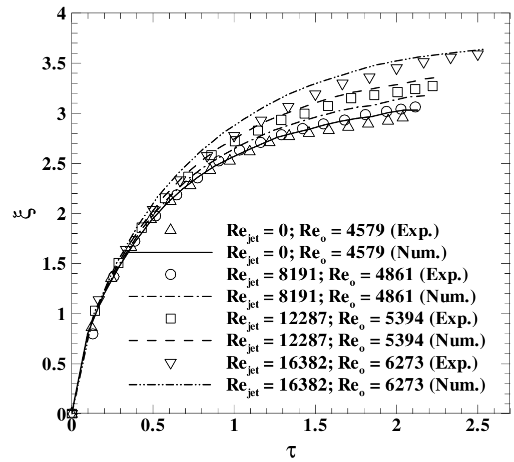

Fig. 8 Spreading factor as a function of time (Propelled droplet)

Figure 9 shows the influence of Re_{jet} on the maximum spreading diameter. The benefit of adding a gas stream into the problem is evident, any augmentation in Re_{jet} leads to an increase in the maximum spreading diameter and therefore, in the wetted area, since inertial forces become more important than surface tension and viscous forces.

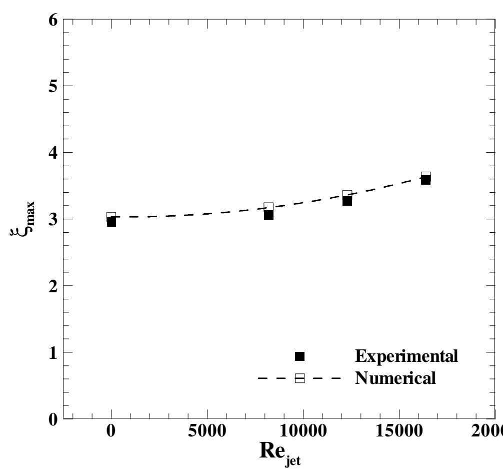

Fig. 9 Maximum spreading factor as a function of Re_{jet}

It is important to understand that the increase in the droplet impact velocity is caused solely by the gas stream, which accelerates the droplet by imposing pressure over its surface. Nevertheless, it is not clear if the gas stream has an important role in the droplet dynamics after the impact. In Fig. 10, a comparison between the streamlines generated in free fall and with the gas stream, for the same droplet impact velocity, is shown.

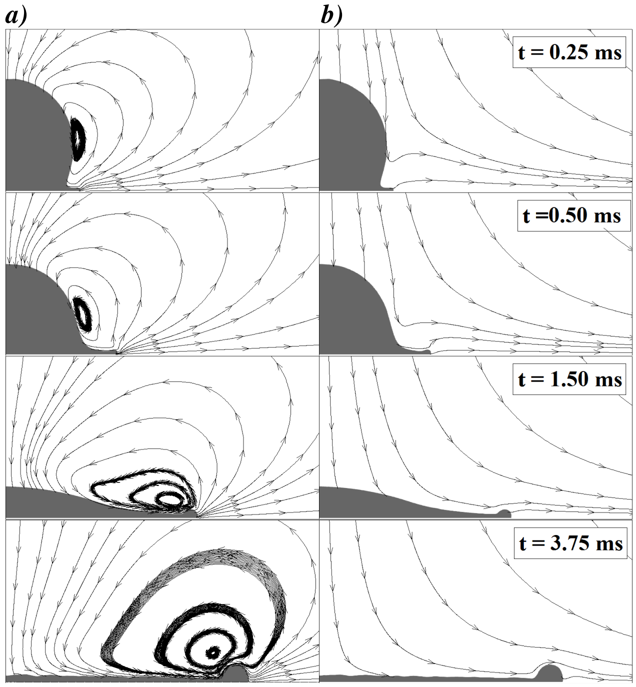

Fig. 10 Streamlines around the droplet for Re_o = 6273 *a)* free fall and *b)* propelled droplet

In free fall (Fig. 10a), since the droplet transports momentum to its surroundings, the droplet generates a recirculation zone whose center is located close to the droplet advancing ring. Thus, the movement of the air is determined by droplet velocity and surface tension at the liquid-gas interface. In contrast, when the gas stream is included (Fig. 10b), the droplet falls into the stagnation region of the jet therefore, no vortex is originated around the droplet while it is spreading. In order to quantify the effect of the imposed pressure and shear stress over the droplet surface by the gas stream during the spreading process, the results were compared against a free falling droplet at the same impact velocities. In Fig. 11, the maximum droplet spreading diameters obtained for a propelled droplet are compared to a droplet in free fall under the same initial conditions.

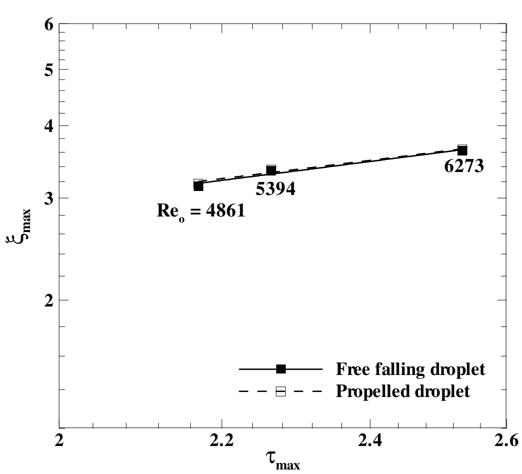

Fig. 11 Maximum droplet spreading factor as a function of non-dimensional time at maximum spreading

For the three cases studied, a slight increase (less than 1%) in maximum droplet spreading diameter was obtained with the inclusion of the gas stream. Hence, the increase in droplet impact velocity appears to be the only important consequence of introducing an air flow, which means that the shear stress that is applied by the gas stream over the droplet surface is not significant enough to produce any important effect on the deformation process.

Figure 12 shows the velocity vectors and the pressure field within the droplet for Re_{jet} = 16382. During the early stage of the impact, it can be observed that the magnitude of the normal component of the velocity is significantly higher than the magnitude of the tangential component. The normal component of the velocity is also higher at the stagnation region and it decreases in the *r* direction. Hence, a large temperature gradient is expected in this direction, where the highest heat transfer would occur at the wall center, where the normal component of the velocity is responsible for the increase in convective heat transfer. In addition, this stage is found to have the highest pressure level. As the water droplet starts to deform, the contact area increases, the pressure becomes more uniform, and the tangential component of the velocity starts to gradually dominate. After maximum spreading is achieved, the inertial force is overcome by the surface tension force and the receding process begins.

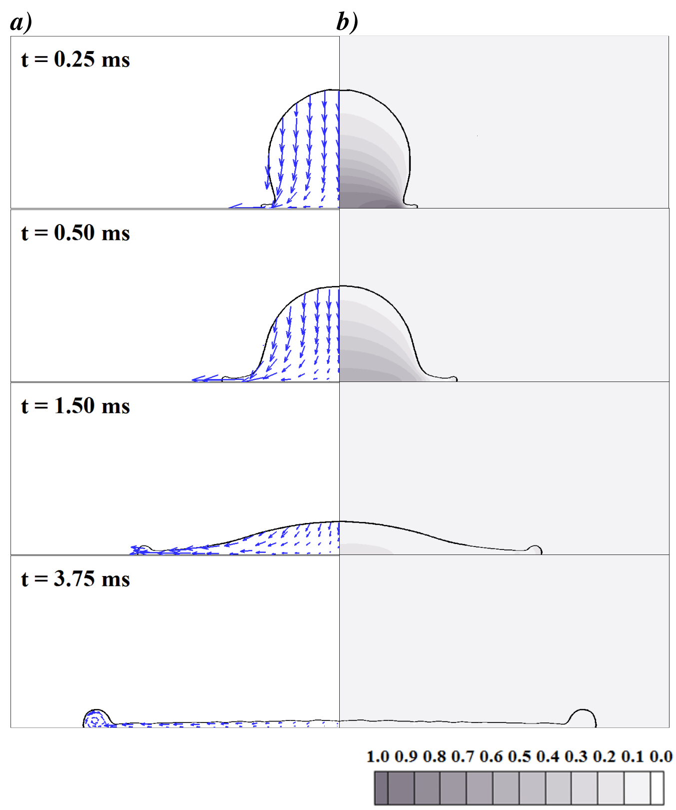

Fig. 12 *a)* Velocity vectors and *b)* pressure field within the droplet for Re_{jet} = 16382

Figure 13 shows the temperature distribution for Re_o = 6273. Figure 13a shows that the region with the highest temperature over the droplet surface corresponds to the region where the air recirculates as a consequence of the droplet motion (Fig. 10a). On the other hand, this region decreases in size when the air flow is introduced (Fig. 13b) until it vanishes completely when the droplet reaches its maximum diameter, since the gas stream carries the hot air away from the droplet surface. Within the droplet, the maximum temperature is achieved inside the advancing ring where the water recirculates due to the surface tension and viscous forces at the liquid-solid interface.

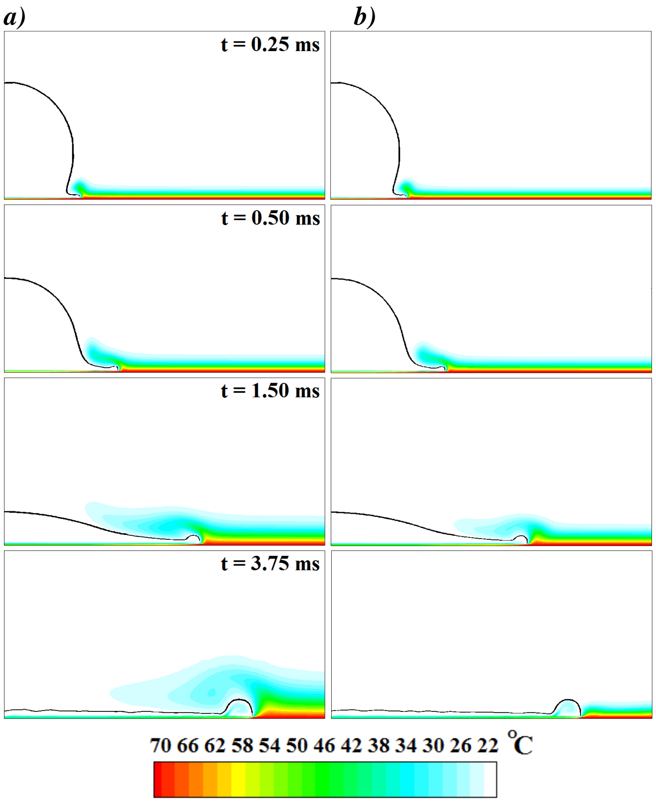

Fig. 13 Temperature distribution comparison between *a)* a free falling droplet and *b)* a propelled droplet (Re_o = 6273)

Figure 14 shows the surface temperature distribution at five different times. As was previously discussed, there is an important temperature gradient in the *r* direction, and this gradient increases as the droplet spreads. Furthermore, there is a rapid change in temperature after the droplet impacts the surface, and this response becomes more significant near the stagnation point. In the stagnation region, the temperature decay appears to be independent of impact velocity, whereas near the advancing ring, the temperature response strongly depends on the impact velocity, which determines the magnitude of the instantaneous droplet spreading diameter. For this reason, in order to measure the cooling effectiveness of a droplet impinging over a heated surface, some previous researchers have used the wetted area. Some definitions of the cooling effectiveness have been given in [30] and [40]. In [30] the cooling effectiveness was defined as the ratio between the heat transfer achieved with and without the presence of the droplet, whereas in [40] the cooling effectiveness was defined as the ratio of the total energy absorbed by the droplet to the total energy that it can absorb.

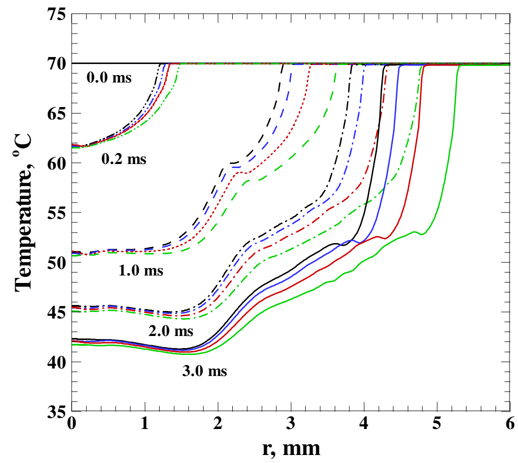

Fig 14 Surface temperature distribution for Re_{jet} = 0 (black line), Re_{jet} = 8191 (blue line), Re_{jet} = 12287 (red line), and Re_{jet} = 16382 (green line)

In this study, the local heat transfer coefficient was calculated with the aim of studying the effects of droplet dynamics in the heat transfer. As shown in Eq. (9), the heat transfer coefficient was defined as the ratio of the instantaneous heat flux that is being transferred from the hot surface to the liquid droplet, to the difference between the wall temperature and the droplet mean temperature

$$h(r,t) = \frac{q_{wall}(r,t)}{T_{wall}(r,t) - \overline{T}_o} \quad (9)$$

The local heat transfer variation is shown in Fig. 15. It is observed that heat transfer in the stagnation region depends mostly on the fluid properties rather than the intensity of the impact velocity. On the other hand, since impact velocity directly affects the spreading rate, there is a strong dependence of the heat transfer coefficient on Re_o in the region closer to the droplet edge.

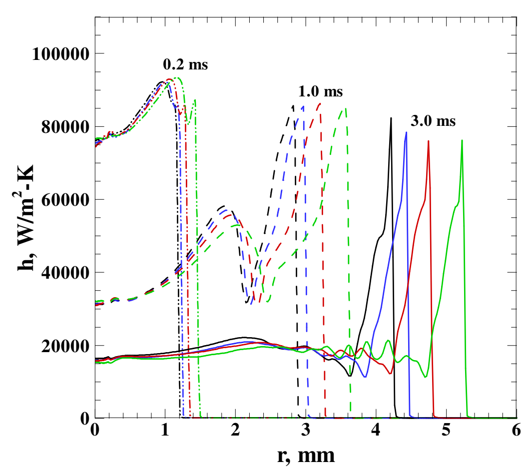

Fig. 15 Surface heat transfer coefficient for Re_{jet} = 0 (black line), Re_{jet} = 8191 (blue line), Re_{jet} = 12287 (red line), and Re_{jet} = 16382 (green line)

Finally, the mean liquid film thickness, normalized by the droplet initial diameter, at the moment of maximum spreading is shown in Fig. 16. As Re_{jet} increases the mean film thickness decreases because of the pressure imposed by the gas stream over the droplet surface, which elevates the impact velocity and thus, the inertial forces that are responsible for the increase in the maximum spreading diameter. It is interesting to note that even when the liquid film thickness decreases by 95% of the initial droplet diameter during the spreading process, its thickness is not small enough to produce a significant effect on surface temperature.

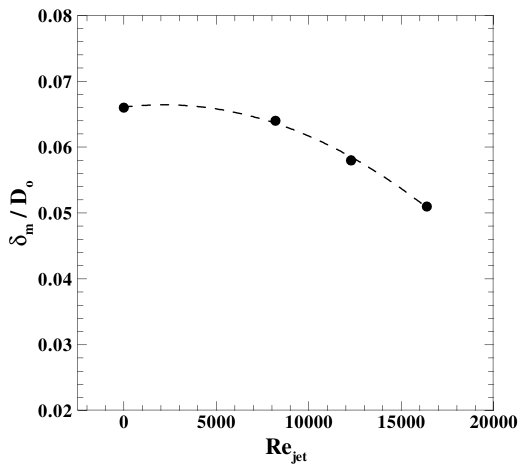

Fig. 16 Normalized film thickness at maximum spreading as a function of Re_{jet}

SUMMARY & CONCLUSIONS

The correlation between droplet dynamics and instantaneous heat transfer during the spreading process of a liquid droplet transported by a gas stream to a target surface was numerically investigated. High-speed photography was used to capture the droplet dynamics throughout the impingement and spreading process, and validate the computational code for the fluid mechanics. Simulations with the VOF model and an adaptive local grid refinement technique yielded good agreement between the experimental and numerical results, with less than 4% difference in the instantaneous droplet spreading diameter. A conjugate solution was found for a stainless steel foil with constant heat generation. The results obtained in free fall were compared to two different problems. The first problem solved the fluid dynamics and heat transfer for a droplet impacting a surface heated with constant heat flux and the second one assumed that the liquid droplet behaves as a semi-infinite solid with constant and uniform heat flux boundary condition. It was found that the advection within the droplet during the early stage of impact has an important effect on the heat transfer through the droplet, which considerably deviates the temperature distribution from the predictions that were obtained considering pure conduction and neglecting the conjugate nature of the problem.

The primary effect of the carrier was to accelerate the droplet by increasing its impact velocity and thus, deforming the droplet more rapidly causing the maximum droplet spreading diameter to increase, and the liquid film thickness to decrease. Therefore, the inclusion of a gas stream led to an increase in wetted area, which enhances the heat transfer from the solid surface to the liquid droplet. Consequently, the radial surface temperature, as well as the heat transfer coefficient, was strongly influenced by the droplet dynamics. In future analysis would be important to consider the entire droplet motion, to examine the correlation between the heat transfer coefficient and liquid film thickness, and its influence in the evaporation rate.

REFERENCES

[1] S. Chandra and C. T. Avedisian, "On the collision of a droplet with a solid surface," Proceedings: Mathematical and Physical Sciences, 432, 13-41, 1991

[2] M. Pasandideh-Fard, Y. M. Qiao, S. Chandra, and J. Mostaghimi, "Capillary effects during droplet impact on a solid surface," Physics of Fluids, 8, 3, 650-659, 1996

[3] X. Zhang and O. A. Basaran, "Dynamic surface tension effects in impact of a drop with a solid surface," Journal of Colloid and Interface Science, 187, 1, 166-178, 1997

[4] M. Bussmann, S. Chandra, and J. Mostaghimi, "Modeling the splash of a droplet impacting a solid surface," Physics of Fluids, 12, 12, 3121-3132, 2000

[5] V. Mehdi-Nejad, J. Mostaghimi, and S. Chandra, "Air bubble entrapment under an impacting droplet," Physics of Fluids, 15, 1, 173-183, 2003

[6] K. K. Haller, Y. Ventikos, D. Poulikakos, and P. Monkewitz, "Computational study of high-speed liquid droplet impact," Journal of Applied Physics, 92, 5, 2821-2828, 2002

[7] S. Sikalo, M. Marengo, C. Tropea, and E. N. Ganic, "Analysis of impact of droplets on horizontal surfaces," Experimental Thermal and Fluid Science, 25, 7, 503-510, 2002

[8] S. Sikalo, H. D. Wilhelm, I. V. Roisman, S. Jakirlic, and C. Tropea, "Dynamic contact angle of spreading droplets: Experiments and simulations," Physics of Fluids, 17, 6, 062103-13, 2005

[9] M. Bussmann, J. Mostaghimi, and S. Chandra, "On a three-dimensional volume tracking model of droplet impact," Physics of Fluids, 11, 6, 1406-1417, 1999

[10] M. Pasandideh-Fard, S. Chandra, and J. Mostaghimi, "A three-dimensional model of droplet impact and solidification," International Journal of Heat and Mass Transfer, 45, 11, 2229-2242, 2002

[11] H. Fujimoto, Y. Shiotani, A. Y. Tong, T. Hama, and H. Takuda, "Three-dimensional numerical analysis of the deformation behavior of droplets impinging onto a solid substrate," International Journal of Multiphase Flow, 33, 3, 317-332, 2007

[12] A. Diaz, A. Ortega, and R. Anderson, "Numerical investigation of a liquid droplet impinging on a heated surface," Lake Buena Vista, Florida, USA, IMECE, 2009

[13] I. V. Roisman, R. Rioboo, and T. Cameron, "Normal impact of a liquid drop on a dry surface: Model for spreading and receding," Proceedings: Mathematical, Physical and Engineering Sciences, 458, 2022, 1411-1430, 2002
[14] L. M. Hocking and A. D. Rivers, "The spreading of a drop by capillary action," Journal of Fluid Mechanics Digital Archive, 121, -1, 425-442, 1982
[15] P. J. Haley and M. J. Miksis, "The effect of the contact line on droplet spreading," Journal of Fluid Mechanics Digital Archive, 223, -1, 57-81, 1991
[16] J. Fukai, M. Tanaka, and O. Miyatake, "Maximum spreading of liquid droplets colliding with flat surfaces," Journal of Chemical Engineering of Japan, 31, 3, 456-461, 1998
[17] P. H. Pidsley, *A numerical investigation of water drop impact*, Storming Media, 1983,
[18] J. Fukai, Z. Zhao, D. Poulikakos, C. M. Megaridis, and O. Miyatake, "Modeling of the deformation of a liquid droplet impinging upon a flat surface," Physics of Fluids A: Fluid Dynamics, 5, 11, 2588-2599, 1993
[19] N. Hatta, H. Fujimoto, and H. Takuda, "Deformation process of a water droplet impinging on a solid surface," Journal of Fluids Engineering, 117, 3, 394-401, 1995
[20] F. H. Harlow and J. P. Shannon, "The splash of a liquid drop," Journal of Applied Physics, 38, 10, 3855-3866, 1967
[21] R. G. Prashant, V. R. Vivek, and V. C. Raghunath, "Dynamics of drop impact on solid surface: Experiments and VOF simulations," AIChE Journal, 51, 1, 59-78, 2005
[22] S. Kamnis and S. Gu, "Numerical modelling of droplet impingement," Journal of Physics D: Applied Physics, 38, 19, 3664-3673, 2005
[23] S. Chandra, M. di Marzo, Y. M. Qiao, and P. Tartarini, "Effect of liquid-solid contact angle on droplet evaporation," Fire Safety Journal, 27, 141-158, 1996
[24] E. F. Crafton and W. Z. Black, "Heat transfer and evaporation rates of small liquid droplets on heated horizontal surfaces," International Journal of Heat and Mass Transfer, 47, 6-7, 1187-1200, 2004
[25] M. di Marzo, P. Tartarini, Y. Liao, D. Evans, and H. Baum, "Evaporative cooling due to a gently deposited droplet," International Journal of Heat and Mass Transfer, 36, 17, 4133-4139, 1993
[26] O. E. Ruiz and W. Z. Black, "Evaporation of water droplets placed on a heated horizontal surface," Journal of Heat Transfer, 124, 5, 854-863, 2002
[27] C. Sodtke, V. Ajaev, and P. Stephan, "Evaporation of thin liquid droplets on heated surfaces," Heat and Mass Transfer, 43, 7, 649-657, 2007
[28] M. di Marzo, P. Tartarini, Y. Liao, D. Evans, and H. Baum, *Dropwise evaporative cooling*, ASME-HTD, 1991, 51-58
[29] N. Nikolopoulos, A. Theodorakakos, and G. Bergeles, "A numerical investigation of the evaporation process of a liquid droplet impinging onto a hot substrate," International Journal of Heat and Mass Transfer, 50, 1-2, 303-319, 2007
[30] G. Strotos, M. Gavaises, A. Theodorakakos, and G. Bergeles, "Numerical investigation of the cooling effectiveness of a droplet impinging on a heated surface," International Journal of Heat and Mass Transfer, 51, 19-20, 4728-4742, 2008
[31] P. Tartarini and M. dl Marzo, "Dropwise evaporative cooling in radiative field," Heat Transfer 1994: Heat transfer augmentation; Two-phase flow with and without phase change; Conduction and insulation, 277, 1994
[32] Y. A. Buevich and V. N. Mankevich, "Impact of an evaporating drop on a heated wall," Journal of Engineering Physics and Thermophysics, 43, 6, 1362-1369, 1982
[33] V. G. Labeish and A. G. Pimenov, "Experimental study of heat transfer between a hot wall and impinging drops," Journal of Engineering Physics and Thermophysics, 47, 6, 1400-1406, 1984
[34] C. O. Pedersen, "An experimental study of the dynamic behavior and heat transfer characteristics of water droplets impinging upon a heated surface," Int. J. Heat Mass Transfer, 13, 369-381, 1970
[35] M. L. Sawyer, S. M. Jeter, and S. I. Abdel-Khalik, "A critical heat flux correlation for droplet impact cooling," International Journal of Heat and Mass Transfer, 40, 9, 2123-2131, 1997
[36] K. K. Tio and S. S. Sadhal, "Thermal analysis of droplet spray evaporation from a heated solid surface," Journal of Heat Transfer (Transactions of the ASME, 1992
[37] L. H. J. Wachters and N. A. J. Westerling, "The heat transfer from a hot wall to impinging water drops in the spheroidal state," Chemical Engineering Science, 21, 11, 1047-1056, 1966
[38] W. M. Healy, J. G. Hartley, and S. I. Abdel-Khalik, "On the validity of the adiabatic spreading assumption in droplet impact cooling," International Journal of Heat and Mass Transfer, 44, 20, 3869-3881, 2001
[39] A. Theodorakakos and G. Bergeles, "Simulation of sharp gas-liquid interface using VOF method and adaptive grid local refinement around the interface," International Journal for Numerical Methods in Fluids, 45, 4, 421-439, 2004
[40] M. Pasandideh-Fard, S. D. Aziz, S. Chandra, and J. Mostaghimi, "Cooling effectiveness of a water drop impinging on a hot surface," International Journal of Heat and Fluid Flow, 22, 2, 201-210, 2001
[41] R. Anderson, A. Diaz, and A. Ortega, "Observations on the behavior of a droplet transported by a gas stream impinging on a isothermal surface," Abu Dhabi, UAE, ICTEA, 2009